LAB WEEK-6

Sort Array by using quick sort

i) First Element as pivot

ii) Last as pivot (iii) Random

[157, 110, 147, 122, 111, 149, 151, 141, 123, 112, 117, 133]

Pivot

(i) 157, 110, 147, 122, 111, 149, 151, 141, 123, 112, 117, 133

Swap

Pivot

S-1 133, 110, 147, 122, 111, 149, 151, 141, 123, 112, 117, 157

Swap

Pivot

S-2 133, 110, 117, 122, 111, 149, 151, 141, 123, 112, 147, 157

Swap

Pivot

S-3 133, 110, 117, 122, 111, ~~149~~ 112, 151, 141, 149 123, 147, 157

Swap

Pivot

S-4 133, 110, 117, 122, 111, 112, 123, 141, 151, 149, 147, 157

Swap

Pivot Pivot

S-5 [123, 110, 117, 122, 111, 112] [133] [141, 151, 149, 147] [157]

Swap

Pivot Pivot

S-6 [112, 110, 117, 122, 111] [123] [133] [141] [151, 149, 147] [157]

Swap Swap

~~110, 117, 112, 122, 111~~

S-7 [112, 110, 111, 122, 117] [123] [133] [141] [147, 149] [151] [157]

Pivot: 112 — Swap

Pivot: 147 — nothing

S-8 [111, 110 [112] 122, 117] [123] [133] [141] [147, 149] [151] [157]

Pivot: 111 — Swap

Pivot: 122 — Swap

S-9 110, 111, 112, 117, 122, 123, 133, 141, 147, 149, 151, 157

9-steps to sort

(ii) St 157, 110, 147, 122, 111, 149, 151, 141, 123, 112, 117, 133

Pivot: 133

Swap

S-1 [117, 110, 147, 122, 111, 149, 151, 141, 123, 112, 157, 133]

Pivot: 133

Swap

St-2 [117, 110, 112, 122, 111, 149, 151, 141, 123, 147, 157, 133]

Pivot: 133

Swap

S-3 [117, 110, 112, 122, 111, 123, 151, 141, 149, 147, 157, 133]

Pivot: 133

Swap

S-4 [117, 110, 112, 122, 111, 123] [133] [141, 149, 147, 157, 151]

Pivot: 123

Pivot: 151

Swap

S-5 [117, 110, 112, 122, 111] [123] [133] [144, 149, 147] [151] [157]

Pivot: 111   Pivot: 147

Swap   swap

S-6 [110, 117, 112, 122, 111] [123] [133] [144] [147] [149] [151] [157]

Pivot: 111

Swap

S-7 [110] [111] [112, 122, 117] [123] [133] [144] [149] [147] [151] [157]

Pivot: 117

swap

S-8 [110] [111] [112] [117] [122] [123] [133] [144] [147] [149] [151] [157]

8-steps to sort

(iii) Random

S-1 157, 110, 147, 122, 111, 149, 151, 141, 123, 112, 117, 133

Pivot: 141

Swap

S-2 133, 110, 147, 122, 111, 149, 151, 141, 123, 112, 117, 157

Pivot: 141

Swap

S-3 133, 110, 117, 122, 111, 149, 151, 141, 123, 112, 147, 157

Pivot: 141

Swap

S-4 133, 110, 117, 122, 111, 112, 151, 141, 123, 149, 147, 157

Pivot: 141

swap

S-5 133, 110, 117, 122, 111, 112, 123, 141, 151, 149, 147, 157

Pivot: 112   Pivot: 147

S-6 133, 110, 117, 122, 111, 112, 123 [141] 151, 149, 147, 157

Swap

S-7 111, 110, 117, 122, 133, 112, 123 [141] [147] 149, 151, 157

Pivot Pivot

S-8 111, 110 [112] [122, 133, 117, 123 [141] [147] 149 [151] [157]

Pivot Pivot Pivot Pivot

S-9 [110] [111] [112] [117] 133, 122, 123 [141] [147] [149] [151] [157]

Pivot

S-10 110, 111, 112, 117, 122, 133, 123, 141, 147, 149, 151, 157

Pivot

Swap

S-11 110, 111, 112, 117, 122, 123, 133, 141, 147, 149, 151, 157

11 steps